

ECE4010J Probabilistic Methods in Engineering (Summer 2026)

Assignment 7

Date Due: See canvas

This assignment has a total of **(85 points)**.

Note: Unless specified otherwise, you must show the details of your work via logical reasoning for each exercise. Simply writing a final result (whether correct or not) will receive **0 point**.

Exercise 7.1 (30 pts) A sample of size 9 from a normal distribution with $\sigma^2 = 25$ is used to test

$$H_0 : \mu = 20$$

$$H_1 : \mu = 28$$

The test statistic used is the sample mean, $\bar{X}$. Let us agree to reject H_0 in favor of H_1 , if the observed value of $\bar{X}$ is greater than 25.

- (i) (5pts) If H_0 is true, what is the distribution of $\bar{X}$?
- (ii) (5pts) Find α . Remember that α is computed under the assumption that H_0 is true.
- (iii) (5pts) If H_1 is true, what is the distribution of $\bar{X}$?
- (iv) (5pts) Find β . Remember that β is computed under the assumption that H_1 is true.
- (v) (5pts) Find the power of the test.
- (vi) (5pts) If the sample size is increased but the critical point is not changed, what will be the effect on α and β ?

Exercise 7.2 (15 pts) Whenever a motorist encounters braking problems, especially an unpredictable pulling to one side, the villain is always held to be the brake pad. Trace elements, especially titanium, can combine with other elements to form minute particles of titanium carbonitride which alter the degree of friction between the pad and disc and lead to unequal wear. The percentage of titanium in a brake pad should not exceed 5%. A study is conducted to detect a situation in which the mean percentage of titanium in the brake pads being produced by a particular manufacturer exceeds 5%.

- (i) (5pts) Set up the appropriate null and alternative hypotheses.
- (ii) (5pts) Discuss the practical consequences of making a Type I and a Type II error.
- (iii) (5pts) A sample of 100 brake pads yields a mean percentage of $\bar{x} = 0.051$. Assume that $\sigma = 0.008$. Find the p -value for the test. Do you think that H_0 should be rejected? Explain. To what type of error are you now subject?

Exercise 7.3 (20 pts) Ozone is a component of smog that can injure sensitive plants even at low levels. In 1979 a federal ozone standard of 0.12 ppm was set. It is thought that the ozone level in air currents over New England exceeds this level. To verify this contention, air samples are obtained from 30 monitoring stations set up across the region.

- (i) (5pts) Set up the appropriate null and alternative hypotheses for verifying the contention.
- (ii) (5pts) What is the critical point for an $\alpha = 0.01$ level test based on a sample of size 30?
- (iii) (5pts) When the data are analyzed, a sample mean of 0.135 and a sample standard deviation of 0.03 are obtained. Use these data to test H_0 . Can H_0 be rejected at the $\alpha = 0.01$ level? What does this mean in a practical sense?
- (iv) (5pts) What assumption are you making concerning the distribution of the random variable X , the ozone level in the air?

Exercise 7.4 (10 pts) A new process for producing small precision parts is being studied. The process consists of mixing fine metal powder with a plastic binder, injecting the mixture into a mold, and then removing the binder with a solvent. These data are obtained on parts that should have a 1-inch diameter and whose standard deviation should not exceed 0.0025 inch:

1.0030	0.9997	0.9990	1.0054	0.9991
1.0041	0.9988	1.0026	1.0032	0.9943
1.0021	1.0028	1.0002	0.9984	0.9999

For these data $\bar{x} = 1.00084$ and $s = 0.00282$.

(i) (5 pts) Test

$$H_0 : \mu = 1, \quad H_1 : \mu \neq 1$$

at the $\alpha = 0.05$ level.

(ii) (5 pts) Test

$$H_0 : \sigma = 0.0025, \quad H_1 : \sigma > 0.0025$$

at the $\alpha = 0.05$ level.

Exercise 7.5 (10 pts) Even with careful workmanship, digital scales may need some adjustment before being put into use. Unless there are systematic errors being made, the apparent zero of the scales before adjustment should fluctuate about true zero. That is, some scales should weigh a little heavy, whereas others should give readings that are a little light. Ten such scales are randomly selected and tested. These data are obtained on the accuracy of the zero reading:

heavy	light	heavy	heavy	heavy
light	light	light	heavy	heavy

Based on these data, can we reject $H_0 : M = 0$ in favor of $H_1 : M > 0$ at the $\alpha = 0.05$ level?